- The frequency, **f** is equal to the number of repetitions per unit of time, and can be calculated from the period, **T**: $\mathbf{f}(\text{Hz}) = \frac{1}{\mathbf{T}(\text{sec})}$ or $\mathbf{T}(\text{sec}) = \frac{1}{\mathbf{f}(\text{Hz})}$.
  When frequency is given in radians/sec then the symbol, **ω** is used, where **ω** = 2π**f**
- Phase Shift, **Θ** of one sinusoidal signal with respect to another (of the same frequency) occurs when there is time-offset, **ΔT** between them. Phase-Shift => $\boldsymbol{\theta} = \frac{\Delta T}{T}.2\pi\ (\text{radians}) = \frac{\Delta T}{T}.360°(\text{degrees})$
- Sinusoidal AC voltage as a function of time: $v(t) = \mathbf{V_P}\cos(\boldsymbol{\omega}t + \boldsymbol{\theta}) = \mathbf{V_P}\cos(2\pi\mathbf{f}t + \boldsymbol{\theta})$

2.0 OBJECTIVES:

- To use Multisim circuit simulation software to plot the step response of second-order dynamic circuits.
- To use the oscilloscope to display the step response of second-order dynamic circuits.
- To measure the parameters that characterize the step response of second-order dynamic circuits.

3.0 REQUIRED LAB EQUIPMENT & PARTS:

- Function Generator (FG) and Oscilloscope.
- ELE202 Lab Kit and ELE302 Lab Kit: various components, breadboard, wires and jumpers.

4.0 PRE-LAB ASSIGNMENT (3 marks with 0.75 marks for each step):

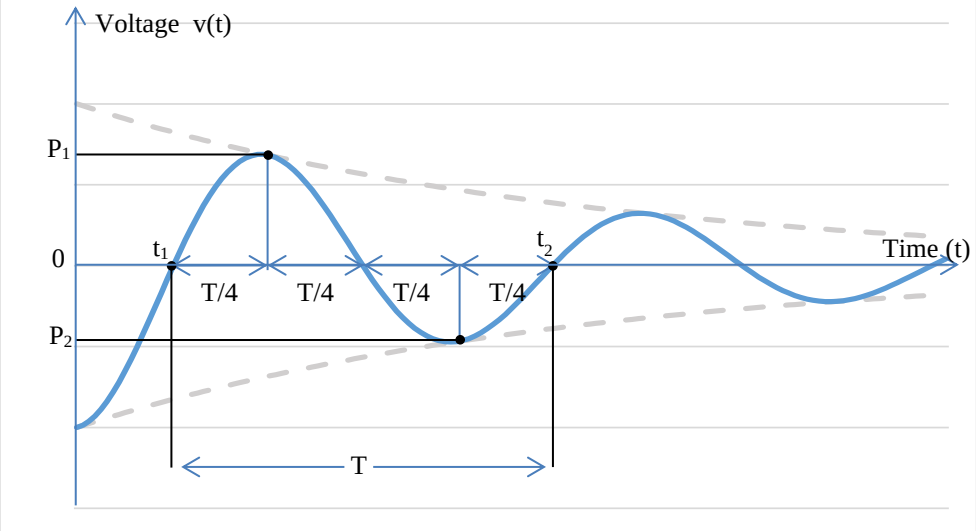


Figure 2.0: An Exponentially Decaying Sinusoidal Function

(a) Step 1: The circuit shown in **Figure 2.0** is an exponentially decaying sinusoidal function that can be characterized by the function: $v(t) = Ae^{-\sigma t}\cos(\boldsymbol{\omega}t + \boldsymbol{\theta})$. The figure defines three variables T, P_1, and P_2. t_1 is the time of the first zero-crossing of v(t) and t_2 is time for the third zero-crossing of v(t). T is the time difference between t_1 and t_2. P_1 is the value of v(t) one quarter of the way between the first zero-crossing and the third zero-crossing (i.e. $P_1 = v(t_1 + \frac{T}{4})$). Finally, P_2 is the value of v(t) three quarters of the way between the first zero-crossing and the third zero-crossing (i.e. $P_2 = v(t_1 + 3\frac{T}{4})$).

i) Find an expression for ω in terms of T.

Pre-Lab workspace (show your analysis here)

Since t_3 is the third zero crossing while t_1 is the first, t_1 and t_3 are one period apart. That means T is exactly 1 period of the cosine function. Since $T = 2\pi w$, $\omega = \frac{2\pi}{T}$

ii) Find an expression for σ in terms of P_1, P_2, and T.

Pre-Lab workspace (show your analysis here)

$P_1 = Ae^{-\sigma(t_1 + T/4)}(1)$ → as the function peaks

$P_2 = Ae^{-\sigma(t_1 + 3/4T)}(-1)$ → as function troughs

$$\frac{P_1}{P_2} = \frac{Ae^{-\sigma(t_1+T/4)}}{-Ae^{-\sigma(t_1+3T/4)}}$$

$$\frac{P_1}{P_2} = -e^{-\sigma[t_1+T/4-(t_1+3T/4)]}$$

$$\frac{P_1}{P_2} = -e^{\sigma(T/2)}$$

$$\sigma\left(\frac{T}{2}\right) = \ln\left(\frac{-P_1}{P_2}\right)$$

$$\sigma = \frac{2}{T}\ln\left|\frac{P_1}{P_2}\right|$$

iii) Find an expression of θ in terms of t_1 and T.

Pre-Lab workspace (show your analysis here)

@ t_1, cosine is 0 while increasing. ∴ the terms inside the cosine is equivilent to $-\frac{\pi}{2}$.

$$\omega t_1 + \theta = -\frac{\pi}{2}$$

$$\theta = -\frac{\pi}{2} - \omega t_1$$

$$= -\frac{\pi}{2} - \frac{2\pi}{T}t_1$$

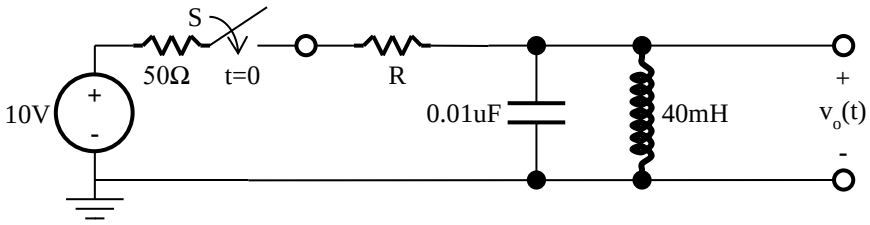
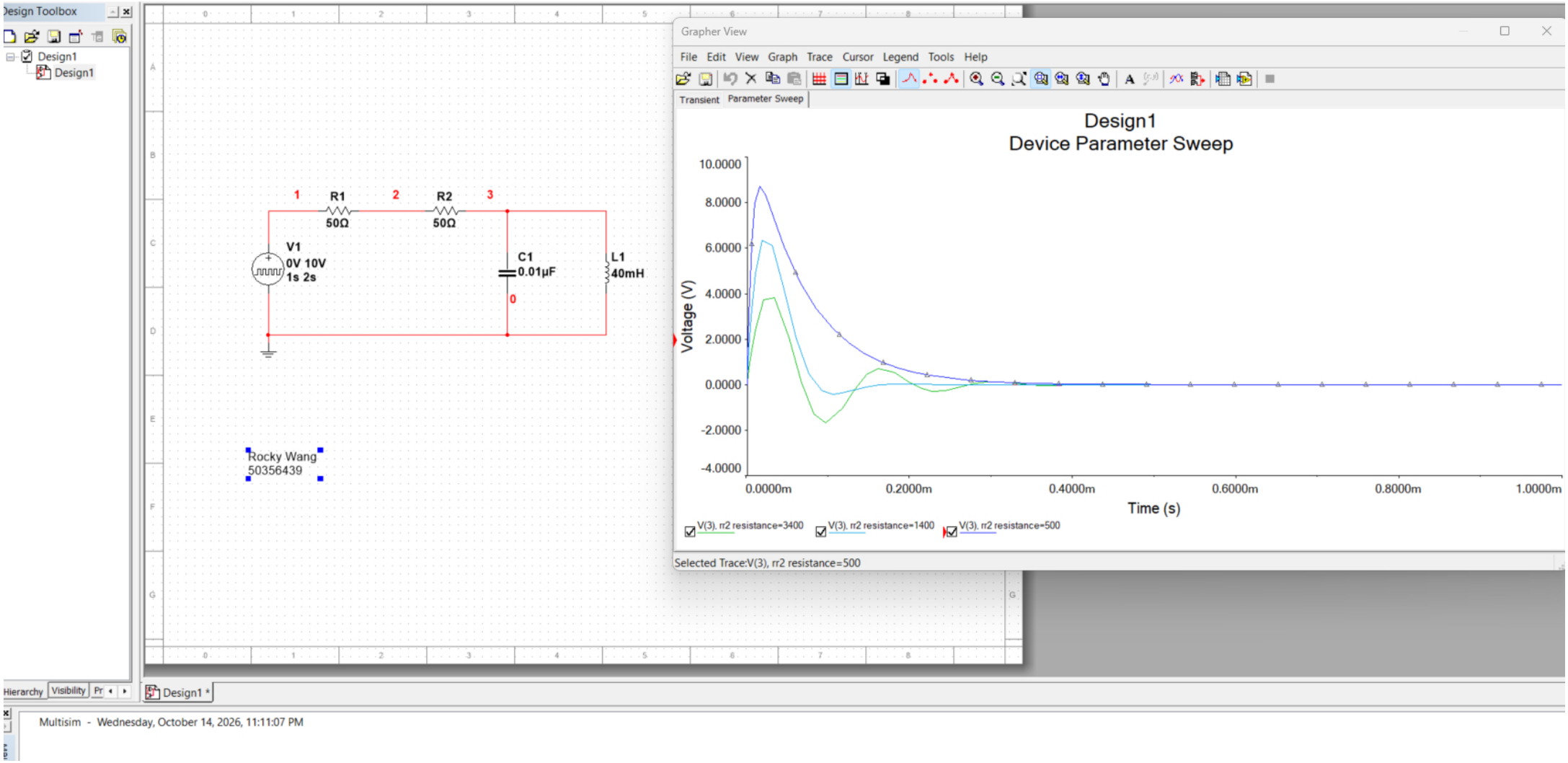


Figure 2.0: Second-Order Bandpass Circuit

(b) Step 2: Consider the dynamic circuit shown in **Figure 2.0**. The switch S has been open for a long time. At t=0, the switch is closed, where it remains for a long time.

i) Use Multisim to plot the step response [$v_o(t)$ for $t \geq 0^+$] when R=1.4kΩ, 3.4kΩ, and 500Ω.

Pre-Lab workspace (show your analysis here)


Design Toolbox
Design1
Design1
1
R1
50Ω
2
R2
50Ω
3
V1
0V 10V
1s 2s
C1
0.01µF
L1
40mH
0
Rocky Wang
50356439
Grapher View
File Edit View Graph Trace Cursor Legend Tools Help
Transient
Parameter Sweep
Design1
Device Parameter Sweep
Voltage (V)
10.0000
8.0000
6.0000
4.0000
2.0000
0.0000
-2.0000
-4.0000
0.0000m
0.2000m
0.4000m
0.6000m
0.8000m
1.0000m
Time (s)
V(3), rr2 resistance=3400
V(3), rr2 resistance=1400
V(3), rr2 resistance=500
Selected Trace:V(3), rr2 resistance=500
Hierarchy
Visibility
Design1 *
Multisim - Wednesday, October 14, 2026, 11:11:07 PM

ii) Use the plots to calculate the parameters (σ and ω) that characterize the step response as: $\mathbf{v_o}(t) = A\boldsymbol{e}^{-\sigma t}\sin(\omega t)$ for R=3.4kΩ.

Pre-Lab workspace (show your analysis here)

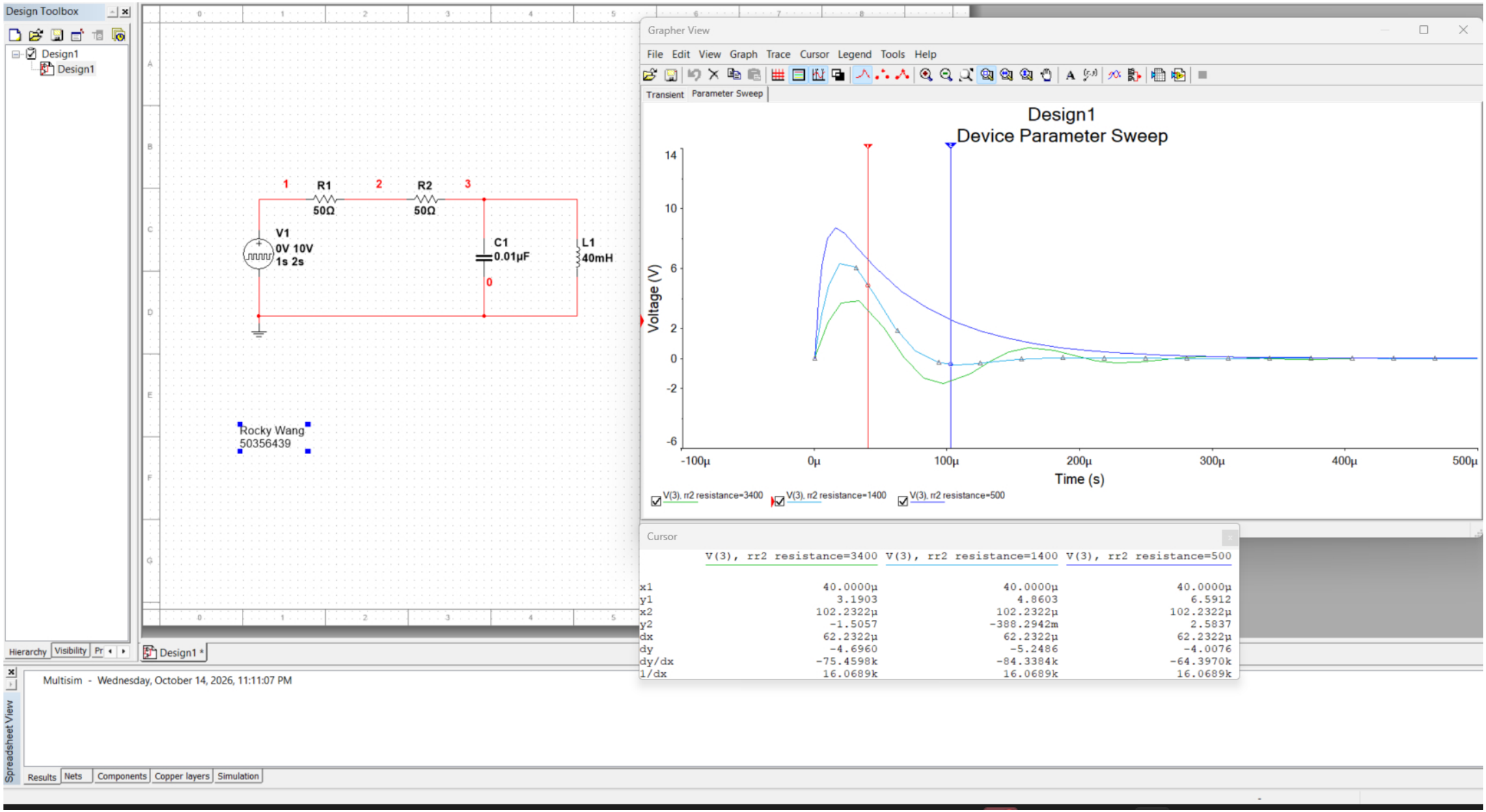
$\Delta x = 136.3\,\mu s = T$ | $t_1 = 0s$ | $P_1 = 3.77\,V$

$T/4 = 34.0774\,\mu s$ | $t_1 + T/4 = 34.0774\,\mu s$ | $P_2 = -1.5057\,V$

$\frac{3}{4}T = 102.2322\,\mu s$ | $t_1 + \frac{3}{4}T = 102.2322\,\mu s$

$$\sigma = \left(\frac{2}{136.3\mu}\right)\ln\left(\frac{3.77}{1.5057}\right) = 13470/s$$

$$\omega = \frac{2\pi}{136.3\mu} = 46100\ rad/s$$


Design Toolbox
Design1
Design1
1
R1
50Ω
2
R2
50Ω
3
V1
0V 10V
1s 2s
C1
0.01µF
0
L1
40mH
Rocky Wang
50356439
Design1 *
Grapher View
File Edit View Graph Trace Cursor Legend Tools Help
Transient Parameter Sweep
Design1
Device Parameter Sweep
Voltage (V)
Time (s)
V(3), rr2 resistance=3400
V(3), rr2 resistance=1400
V(3), rr2 resistance=500
Cursor
V(3), rr2 resistance=3400 V(3), rr2 resistance=1400 V(3), rr2 resistance=500
x1 40.0000µ 40.0000µ 40.0000µ
y1 3.1903 4.8603 6.5912
x2 102.2322µ 102.2322µ 102.2322µ
y2 -1.5057 -388.2942m 2.5837
dx 62.2322µ 62.2322µ 62.2322µ
dy -4.6960 -5.2486 -4.0076
dy/dx -75.4598k -84.3384k -64.3970k
1/dx 16.0689k 16.0689k 16.0689k
Multisim - Wednesday, October 14, 2026, 11:11:07 PM
Spreadsheet View
Results Nets Components Copper layers Simulation

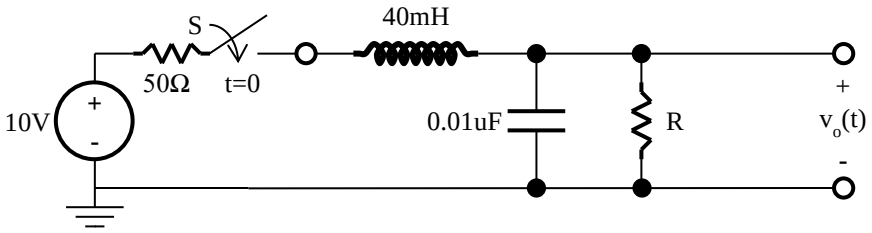
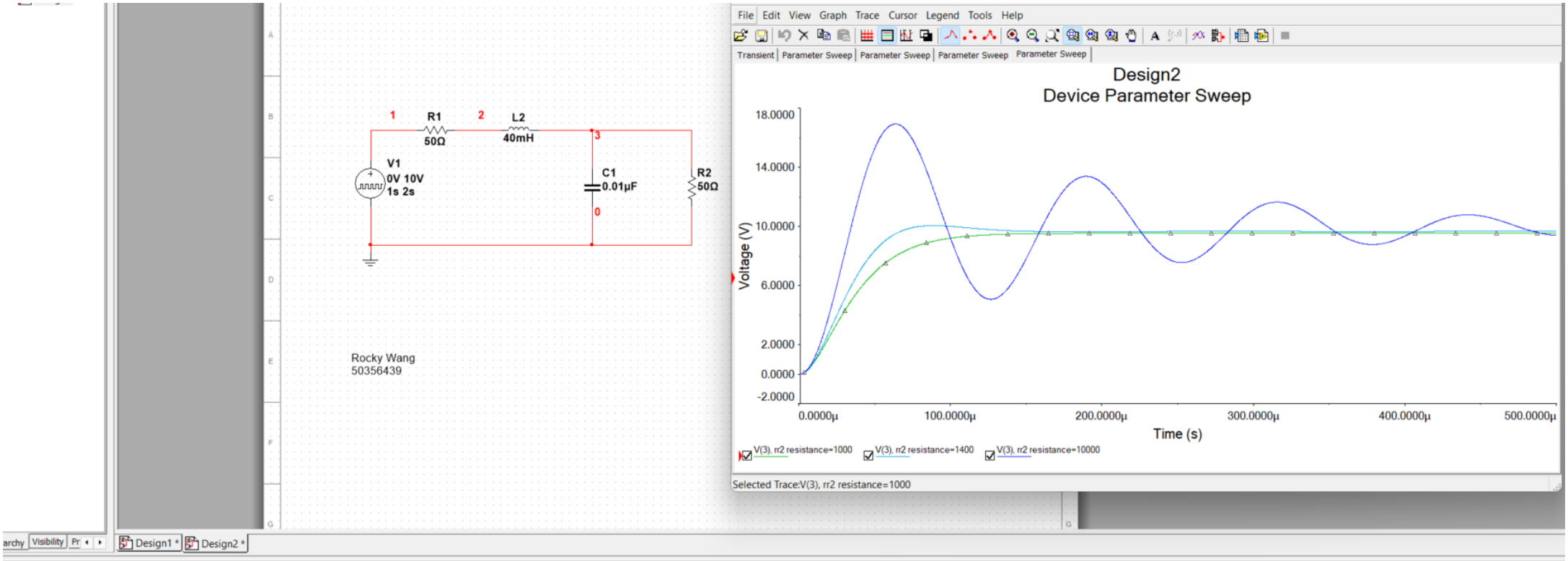


Figure 3.0: Second-Order Lowpass Circuit

(c) Step 3: Consider the dynamic circuit shown in **Figure 3.0.** The switch S has been open for a long time. At t=0, the switch is closed, where it remains for a long time.

i) Use Multisim to plot the step response [$v_o(t)$ for $t \geq 0^+$] when R=1kΩ, 1.4kΩ, and 10kΩ.

Pre-Lab workspace (show your analysis here)



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ii) Use the plots to calculate the parameters (σ and ω) that characterize the step response as: $\mathbf{v_o}(t) = B + Ae^{-\sigma t}\cos(\omega t + \theta)$ for R=10kΩ.

Pre-Lab workspace (show your analysis here)

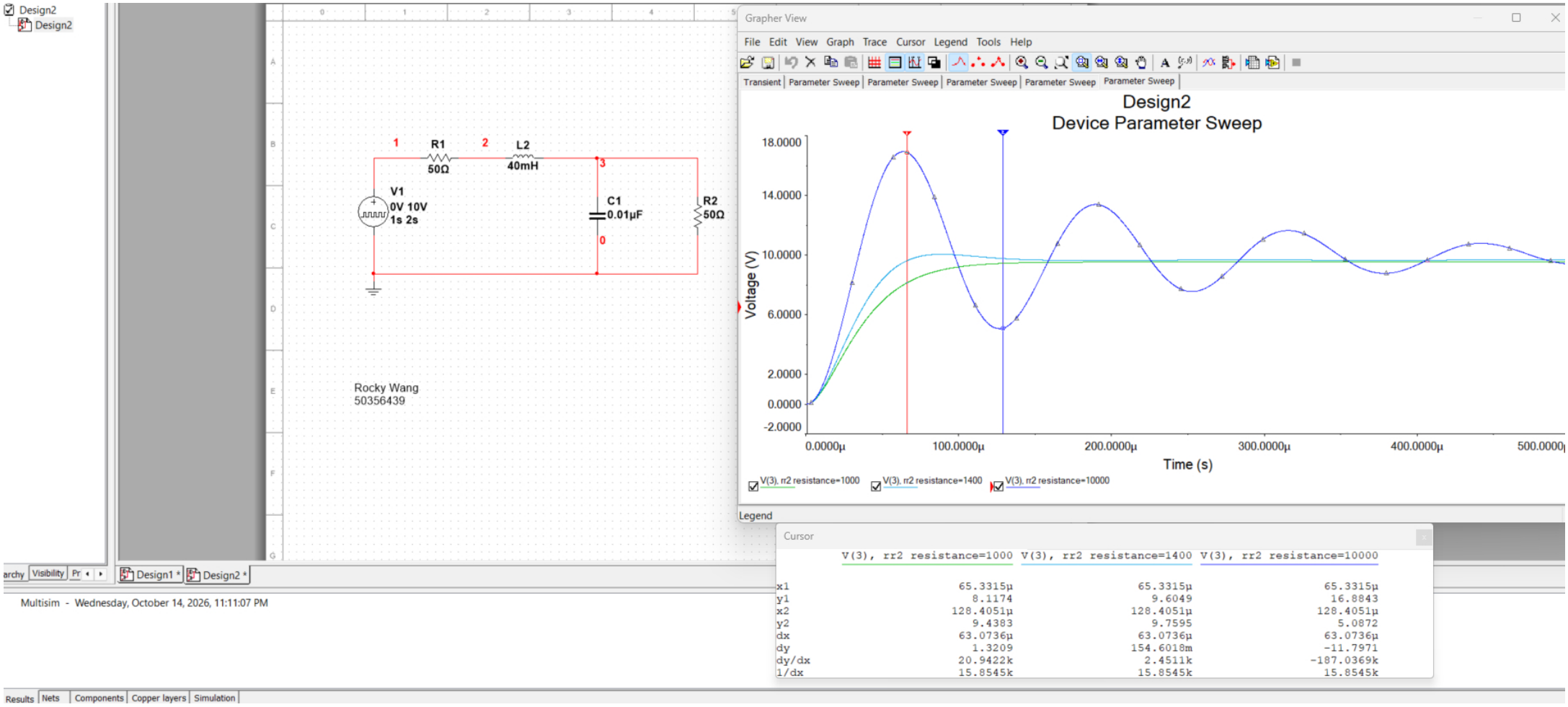
$\Delta x = 126.1472\,\mu s = T$ | $t_1 = 33.7947\,\mu s$ | $P_1 = 16.8843 - 9.95 = 6.934\,V$

$T/4 = 31.5368\,\mu s$ | $t_1 + T/4 = 65.3315\,\mu s$ | $P_2 = 5.0872 - 9.95 = -4.863\,V$

$\frac{3}{4}T = 94.6104\,\mu s$ | $t_1 + \frac{3}{4}T = 128.4051\,\mu s$

$$\sigma = \left(\frac{2}{126.1472\mu}\right)\ln\left(\frac{6.934}{4.863}\right) = 5626/s$$

$$\omega = \frac{2\pi}{126.1472\mu} = 49810\ rad/s$$


Design2
Design2
1
R1
50Ω
2
L2
40mH
3
V1
0V 10V
1s 2s
C1
0.01µF
R2
50Ω
0
Rocky Wang
50356439
Grapher View
File Edit View Graph Trace Cursor Legend Tools Help
Transient | Parameter Sweep | Parameter Sweep | Parameter Sweep | Parameter Sweep | Parameter Sweep
Design2
Device Parameter Sweep
Voltage (V)
18.0000
14.0000
10.0000
6.0000
2.0000
0.0000
-2.0000
0.0000µ
100.0000µ
200.0000µ
300.0000µ
400.0000µ
500.0000
Time (s)
V(3), rr2 resistance=1000
V(3), rr2 resistance=1400
V(3), rr2 resistance=10000
Legend
Cursor
V(3), rr2 resistance=1000 V(3), rr2 resistance=1400 V(3), rr2 resistance=10000
x1 65.3315µ 65.3315µ 65.3315µ
y1 8.1174 9.6049 16.8843
x2 128.4051µ 128.4051µ 128.4051µ
y2 9.4383 9.7595 5.0872
dx 63.0736µ 63.0736µ 63.0736µ
dy 1.3209 154.6018m -11.7971
dy/dx 20.9422k 2.4511k -187.0369k
1/dx 15.8545k 15.8545k 15.8545k
Visibility
Design1 *
Design2 *
Multisim - Wednesday, October 14, 2026, 11:11:07 PM
Results
Nets
Components
Copper layers
Simulation

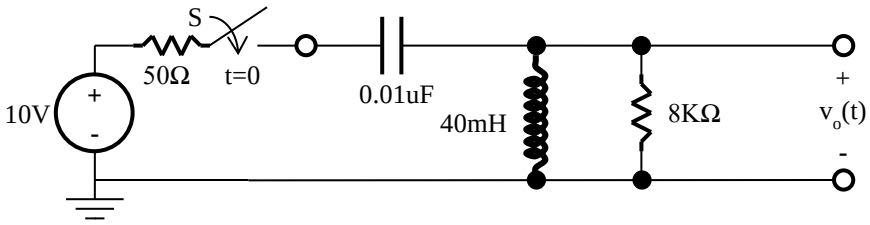
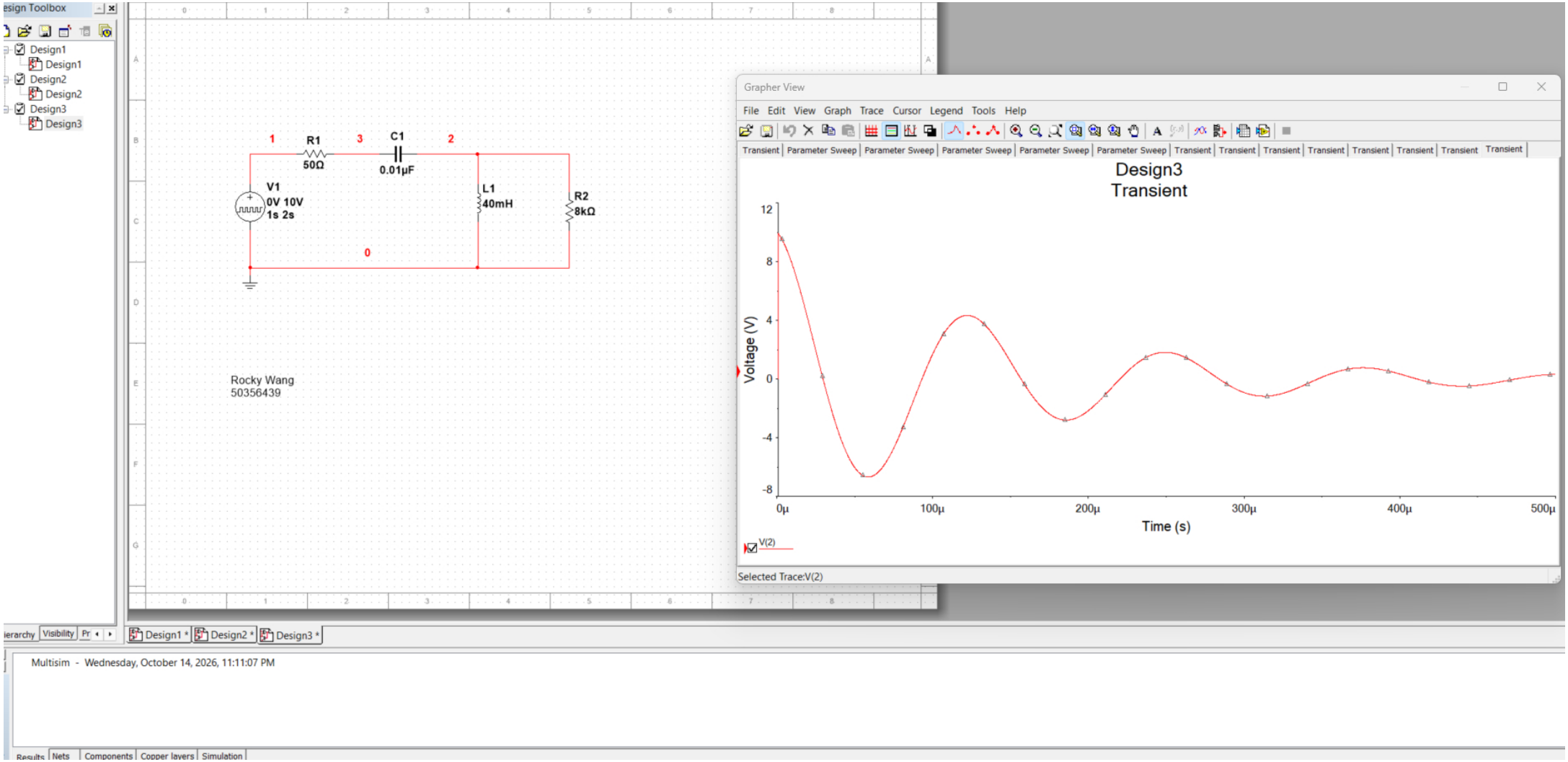


Figure 4.0: Second-Order Highpass Circuit

(d) Step 4: Consider the dynamic circuit shown in **Figure 4.0.** The switch S has been open for a long time. At t=0, the switch is closed, where it remains for a long time.

i) Use Multisim to plot the step response [$v_o(t)$ for $t \geq 0^+$].

Pre-Lab workspace (show your analysis here)


Design Toolbox
Design1
Design1
Design2
Design2
Design3
Design3
1
R1
50Ω
3
C1
0.01µF
2
V1
0V 10V
1s 2s
L1
40mH
R2
8kΩ
0
Rocky Wang
50356439
Grapher View
File Edit View Graph Trace Cursor Legend Tools Help
Design3
Transient
Voltage (V)
Time (s)
12
8
4
0
-4
-8
0µ
100µ
200µ
300µ
400µ
500µ
V(2)
Selected Trace:V(2)
Design1 *
Design2 *
Design3 *
Multisim - Wednesday, October 14, 2026, 11:11:07 PM
Results
Nets
Components
Copper layers
Simulation

ii) Use the plot to calculate the parameters (σ and ω) that characterize the step response as: $\mathbf{v_o}(t) = Ae^{-\sigma t}\cos(\omega t + \theta)$.

Pre-Lab workspace (show your analysis here)

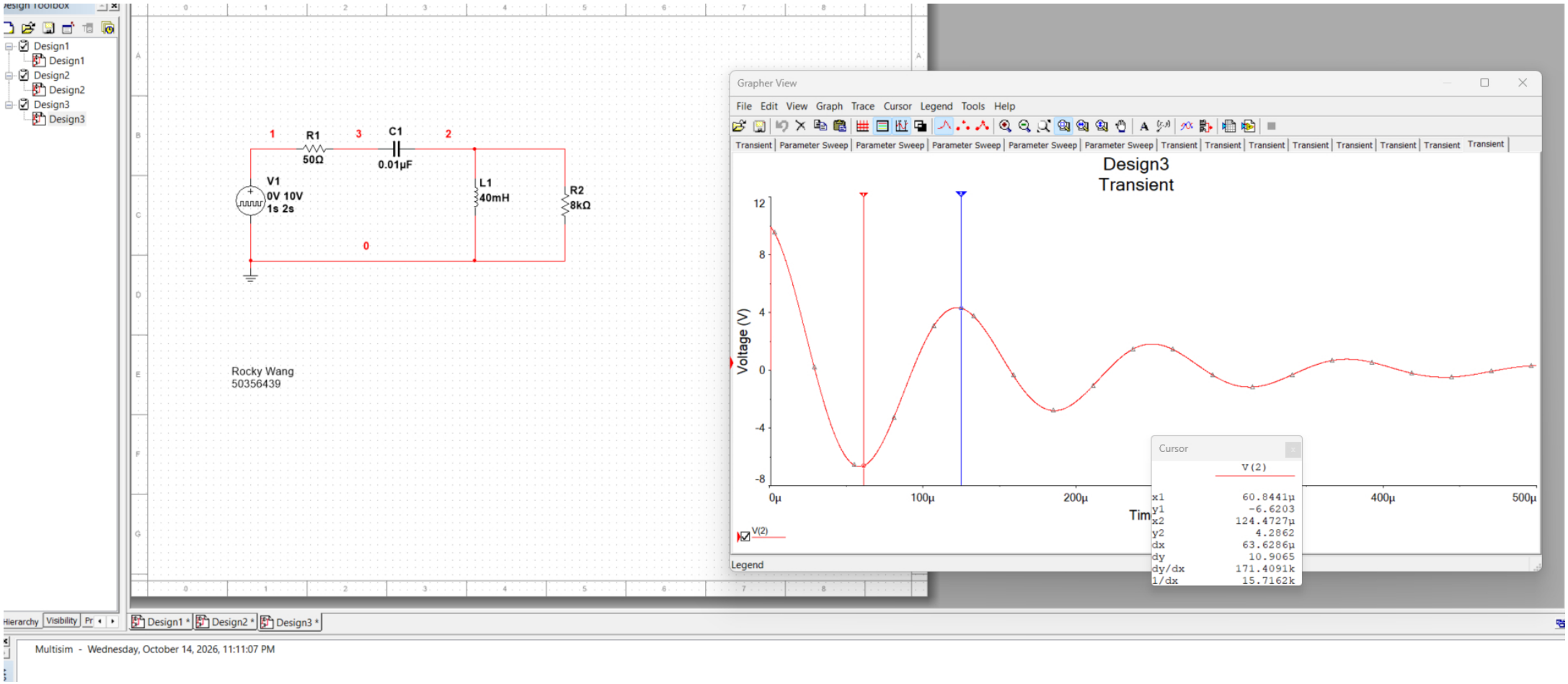
$\Delta x = 127.2572\,\mu s$ | $t_1 = 29.0298\,\mu s$ | $P_1 = -6.6203\,V$

$T/4 = 31.8143\,\mu s$ | $t_1 + T/4 = 60.8441\,\mu s$ | $P_2 = 4.2862\,V$

$3/4\,T = 95.4429\,\mu s$ | $t_1 + 3/4\,T = 124.4727\,\mu s$

$$\sigma = \left(\frac{2}{127.2572\mu}\right)\ln\left(\frac{6.6203}{4.2862}\right) = 6832/s$$

$$\omega = \frac{2\pi}{127.2572\mu} = 49370\ rad/s$$


Design Toolbox
Design1
Design1
Design2
Design2
Design3
Design3
1
R1
50Ω
3
C1
0.01µF
2
V1
0V 10V
1s 2s
L1
40mH
R2
8kΩ
0
Rocky Wang
50356439
Grapher View
File Edit View Graph Trace Cursor Legend Tools Help
Transient | Parameter Sweep | Parameter Sweep | Parameter Sweep | Parameter Sweep | Parameter Sweep | Transient | Transient | Transient | Transient | Transient | Transient | Transient | Transient
Design3
Transient
Voltage (V)
12
8
4
0
-4
-8
0µ
100µ
200µ
400µ
500µ
V(2)
Legend
Cursor
V(2)
x1 60.8441µ
y1 -6.6203
x2 124.4727µ
y2 4.2862
dx 63.6286µ
dy 10.9065
dy/dx 171.4091k
1/dx 15.7162k
Hierarchy
Visibility
Design1 *
Design2 *
Design3 *
Multisim - Wednesday, October 14, 2026, 11:11:07 PM